

Wave 2 Statistical Analyses

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Hierarchical Bayesian Models Wave 2

This document runs the models used to analyse the main outcomes for Wave 2.
The structure of the document follows the Analysis_Plans.txt file.
The variables in the datasets are explained in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.38    0.01 0.36   -0.32    0.14    0.38    0.62    1.08  4876
## sigma        0.88    0.00 0.21    0.55    0.73    0.85    1.00    1.36 11827
## alpha[1]     -0.62    0.01 0.69   -1.99   -1.07   -0.62   -0.18    0.73  6714
## alpha[2]      0.18    0.01 0.73   -1.29   -0.31    0.18    0.66    1.62  8274
## alpha[3]      1.58    0.01 0.71    0.22    1.11    1.56    2.03    3.02  6617
## alpha[4]      0.07    0.00 0.27   -0.46   -0.11    0.07    0.25    0.60  5972
## alpha[5]      0.97    0.01 0.60   -0.19    0.56    0.96    1.36    2.18 11339
## alpha[6]     -0.40    0.01 0.52   -1.46   -0.74   -0.38   -0.04    0.60 10531
## alpha[7]     -0.33    0.00 0.43   -1.18   -0.61   -0.32   -0.04    0.50  8032
## alpha[8]      1.32    0.01 0.48    0.41    1.00    1.31    1.63    2.28  8259
## alpha[9]      0.63    0.00 0.33   -0.01    0.41    0.63    0.85    1.30 10545
## alpha[10]    -0.22    0.00 0.45   -1.12   -0.52   -0.22    0.09    0.64  9306
## alpha[11]     0.43    0.01 0.56   -0.66    0.05    0.42    0.80    1.55 11747
## alpha[12]     1.48    0.01 0.68    0.23    1.02    1.44    1.90    2.93 11901
## alpha[13]     0.42    0.00 0.50   -0.56    0.08    0.41    0.76    1.42 10196
## alpha[14]     0.03    0.01 0.58   -1.12   -0.35    0.03    0.42    1.15 12666
## alpha[15]     1.37    0.01 0.49    0.42    1.03    1.36    1.69    2.35  7660
## alpha[16]    -0.17    0.00 0.56   -1.30   -0.54   -0.16    0.20    0.92 12835
## alpha[17]    -0.32    0.01 0.53   -1.38   -0.66   -0.31    0.04    0.71 10456
## alpha[18]    -0.13    0.00 0.53   -1.20   -0.47   -0.13    0.23    0.89 12378
## alpha[19]     0.94    0.01 0.52   -0.06    0.58    0.93    1.28    1.97  7906
```

```

## beta1      -1.31    0.01 0.84   -3.01   -1.85   -1.30   -0.75    0.30  7254
## beta2      -0.17    0.00 0.15   -0.45   -0.27   -0.17   -0.07    0.12  4456
## beta3       1.36    0.00 0.33    0.73    1.14    1.36    1.58    2.01 10788
## lp__       -411.52   0.05 3.85 -419.98 -413.92 -411.17 -408.77 -404.98 6689
##           Rhat
## mu          1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1
## alpha[10]    1
## alpha[11]    1
## alpha[12]    1
## alpha[13]    1
## alpha[14]    1
## alpha[15]    1
## alpha[16]    1
## alpha[17]    1
## alpha[18]    1
## alpha[19]    1
## beta1        1
## beta2        1
## beta3        1
## lp__         1
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 11:55:36 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

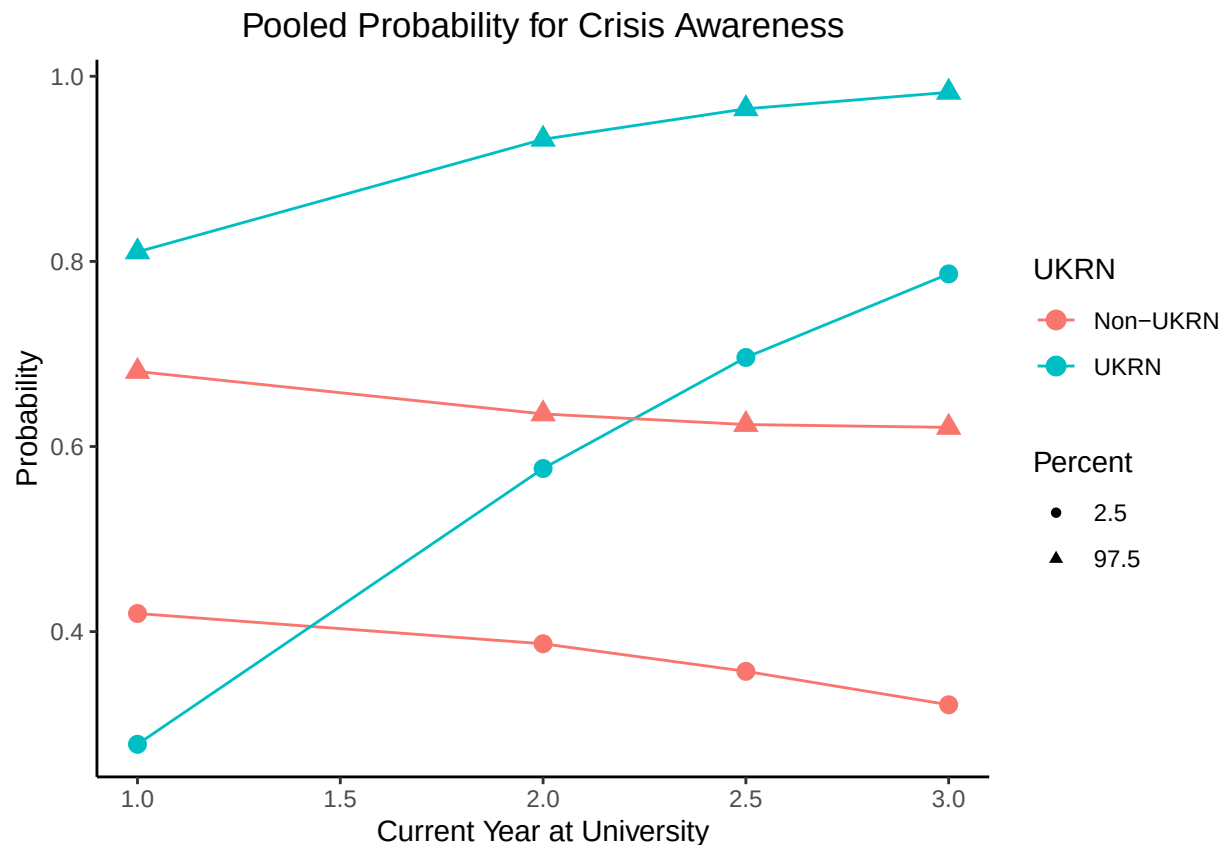
Number of participants in the analysis

Total N = 682
UKRN N = 221
Non-UKRN N = 461
Year 1 N = 308
Year 2 N = 255
Placement Year N = 31
Year 4 N = 88

The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.7311188
 97.5% = 2.0065127

The interpretation is that students who attend a UKRN affiliated university are more likely to be aware of the reproducibility crisis in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



Learned about Open Science at Uni

Bayesian Logistic regression for whether participants received teaching about open science or not (yes/no).
The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement of third year * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean  sd    2.5%    25%    50%    75%   97.5% n_eff
## mu          -1.55    0.01 0.40   -2.36   -1.81   -1.54   -1.28   -0.78  3296
## sigma        0.79    0.00 0.21    0.45    0.64    0.76    0.91    1.27  7620
## alpha[1]    -2.28    0.01 0.66   -3.65   -2.70   -2.26   -1.84   -1.00  4618
## alpha[2]    -1.37    0.01 0.70   -2.75   -1.82   -1.37   -0.91    0.00  5335
## alpha[3]    -0.99    0.01 0.66   -2.30   -1.41   -0.99   -0.56    0.31  4460
## alpha[4]    -0.49    0.00 0.32   -1.11   -0.70   -0.49   -0.27    0.14  4234
## alpha[5]    -1.20    0.01 0.63   -2.45   -1.62   -1.19   -0.77    0.04  7338
## alpha[6]    -2.08    0.01 0.67   -3.49   -2.51   -2.05   -1.62   -0.86  7433
## alpha[7]    -2.06    0.01 0.57   -3.23   -2.43   -2.04   -1.67   -0.99  5977
## alpha[8]    -2.12    0.01 0.60   -3.36   -2.51   -2.09   -1.70   -0.99  6450
## alpha[9]    -2.10    0.01 0.48   -3.10   -2.41   -2.08   -1.77   -1.21  8014
## alpha[10]   -2.04    0.01 0.60   -3.28   -2.42   -2.01   -1.62   -0.95  7012
## alpha[11]   -1.84    0.01 0.69   -3.27   -2.28   -1.82   -1.37   -0.57  8230
```

```

## alpha[12]    -0.77    0.01 0.65   -2.02   -1.21   -0.77   -0.34    0.53  8480
## alpha[13]    -1.21    0.01 0.58   -2.36   -1.60   -1.21   -0.82   -0.09  7128
## alpha[14]    -1.79    0.01 0.69   -3.24   -2.24   -1.77   -1.32   -0.52  8551
## alpha[15]    -1.06    0.01 0.52   -2.08   -1.41   -1.07   -0.72   -0.04  5404
## alpha[16]    -1.88    0.01 0.69   -3.31   -2.32   -1.85   -1.42   -0.62  7700
## alpha[17]    -1.49    0.01 0.60   -2.70   -1.87   -1.48   -1.09   -0.33  6918
## alpha[18]    -1.96    0.01 0.68   -3.39   -2.39   -1.92   -1.50   -0.72  7840
## alpha[19]    -0.66    0.01 0.57   -1.76   -1.05   -0.66   -0.27    0.47  5401
## beta1        -0.12    0.01 0.81   -1.70   -0.65   -0.12    0.41    1.48  5322
## beta2        -0.09    0.00 0.18   -0.44   -0.21   -0.09    0.03    0.25  3337
## beta3         0.74    0.00 0.31    0.14    0.53    0.74    0.95    1.35  6522
## lp__         -362.13   0.06 4.27 -371.31 -364.83 -361.82 -359.08 -354.82 5199
##              Rhat
## mu              1
## sigma           1
## alpha[1]        1
## alpha[2]        1
## alpha[3]        1
## alpha[4]        1
## alpha[5]        1
## alpha[6]        1
## alpha[7]        1
## alpha[8]        1
## alpha[9]        1
## alpha[10]       1
## alpha[11]       1
## alpha[12]       1
## alpha[13]       1
## alpha[14]       1
## alpha[15]       1
## alpha[16]       1
## alpha[17]       1
## alpha[18]       1
## alpha[19]       1
## beta1           1
## beta2           1
## beta3           1
## lp__            1
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 11:59:11 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 679

UKRN N = 220

Non-UKRN N = 459

Year 1 N = 308

Year 2 N = 253

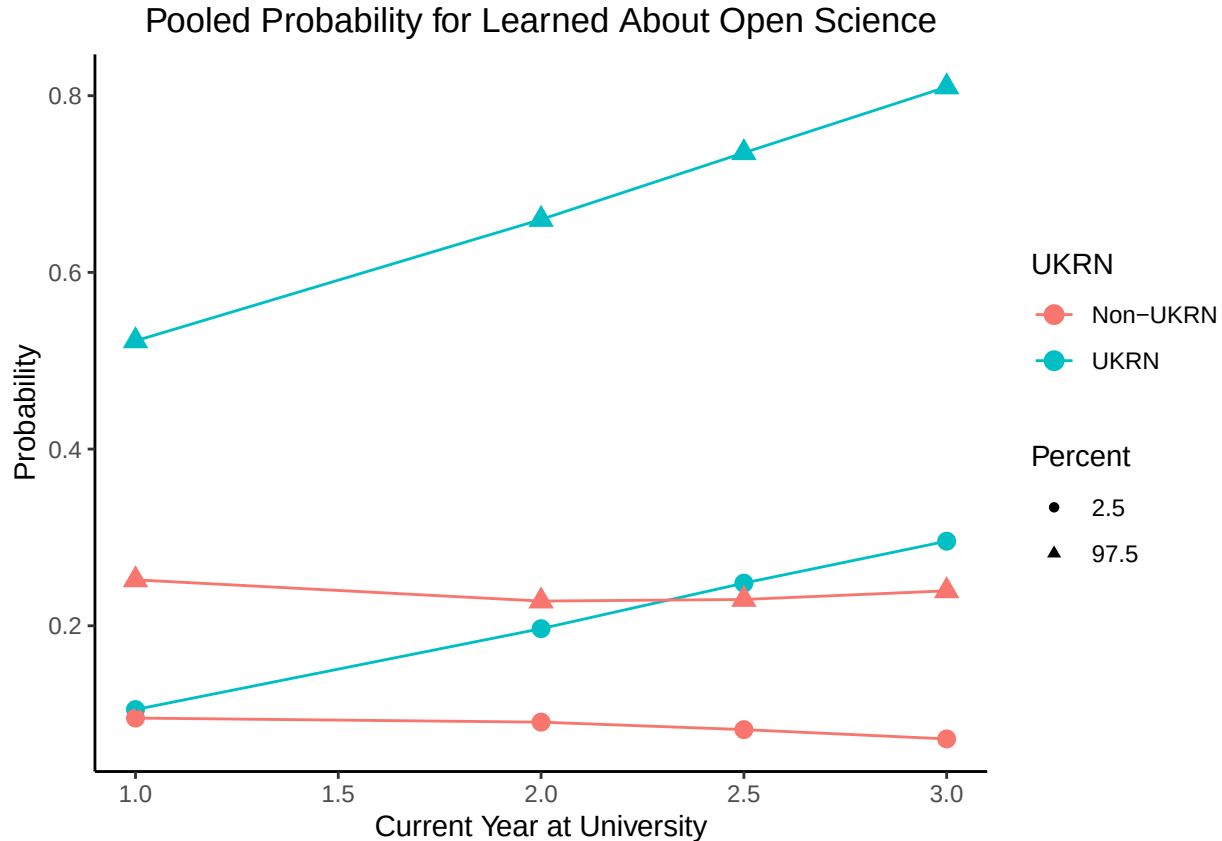
Placement Year N = 31

Year 4 N = 87

The output shows that beta credibility intervals do not cross 0 only for the interaction term.

2.5% = 0.1438867
 97.5% = 1.3533678

The interpretation is that students who attend a UKRN affiliated university are more likely to have learned about open science in the later years of education. For students who attend a non-UKRN affiliated university, there is no difference.



Can explain open science

Bayesian Logistic regression for whether participants can explain what open science is (yes/no).

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement of third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu          -1.23    0.03  0.31   -1.85   -1.43   -1.22   -0.99   -0.67   101
## sigma        0.35    0.03  0.17    0.05    0.25    0.35    0.46    0.71    37
## alpha[1]    -1.31    0.04  0.43   -2.20   -1.60   -1.30   -0.99   -0.55   118
## alpha[2]    -1.08    0.01  0.44   -1.95   -1.36   -1.05   -0.81   -0.19  1656
## alpha[3]    -1.28    0.04  0.42   -2.17   -1.57   -1.26   -0.97   -0.53   141
## alpha[4]    -0.80    0.01  0.27   -1.34   -0.97   -0.82   -0.62   -0.24  1829
## alpha[5]    -1.12    0.02  0.42   -1.99   -1.40   -1.09   -0.83   -0.30   704
## alpha[6]    -1.39    0.04  0.44   -2.33   -1.68   -1.37   -1.04   -0.63   100
```

```

## alpha[7]      -1.37      0.04  0.41  -2.22  -1.65  -1.36  -1.05  -0.67    96
## alpha[8]      -1.31      0.04  0.41  -2.16  -1.58  -1.29  -1.00  -0.58   119
## alpha[9]      -1.49      0.06  0.39  -2.29  -1.76  -1.50  -1.21  -0.81    48
## alpha[10]     -1.45      0.05  0.44  -2.37  -1.74  -1.43  -1.11  -0.74    68
## alpha[11]     -1.32      0.04  0.44  -2.27  -1.61  -1.29  -0.99  -0.56   113
## alpha[12]     -1.05      0.01  0.44  -1.93  -1.34  -1.03  -0.82  -0.16  2536
## alpha[13]     -1.25      0.03  0.42  -2.13  -1.52  -1.22  -0.95  -0.48   201
## alpha[14]     -1.08      0.01  0.42  -1.93  -1.36  -1.06  -0.81  -0.23  1093
## alpha[15]     -1.33      0.04  0.41  -2.18  -1.60  -1.30  -1.01  -0.61   104
## alpha[16]     -1.15      0.02  0.41  -2.00  -1.42  -1.12  -0.88  -0.35   484
## alpha[17]     -1.47      0.06  0.47  -2.48  -1.78  -1.44  -1.11  -0.71    73
## alpha[18]     -1.20      0.02  0.41  -2.06  -1.47  -1.17  -0.92  -0.41   318
## alpha[19]     -0.83      0.01  0.43  -1.64  -1.11  -0.86  -0.57   0.08  3124
## beta1          0.05      0.01  0.57  -1.07  -0.31   0.02   0.41   1.22  4108
## beta2          0.12      0.01  0.14  -0.15   0.02   0.12   0.22   0.41   290
## beta3          0.31      0.01  0.26  -0.20   0.14   0.29   0.48   0.85  2256
## lp__          -403.62     2.43 10.05 -417.51 -409.79 -405.65 -400.64 -375.89    17
##
## Rhat
## mu            1.05
## sigma         1.14
## alpha[1]      1.04
## alpha[2]      1.01
## alpha[3]      1.04
## alpha[4]      1.01
## alpha[5]      1.02
## alpha[6]      1.05
## alpha[7]      1.05
## alpha[8]      1.04
## alpha[9]      1.10
## alpha[10]     1.07
## alpha[11]     1.04
## alpha[12]     1.01
## alpha[13]     1.03
## alpha[14]     1.01
## alpha[15]     1.05
## alpha[16]     1.02
## alpha[17]     1.06
## alpha[18]     1.02
## alpha[19]     1.00
## beta1         1.00
## beta2         1.03
## beta3         1.01
## lp__          1.37
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 12:02:04 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

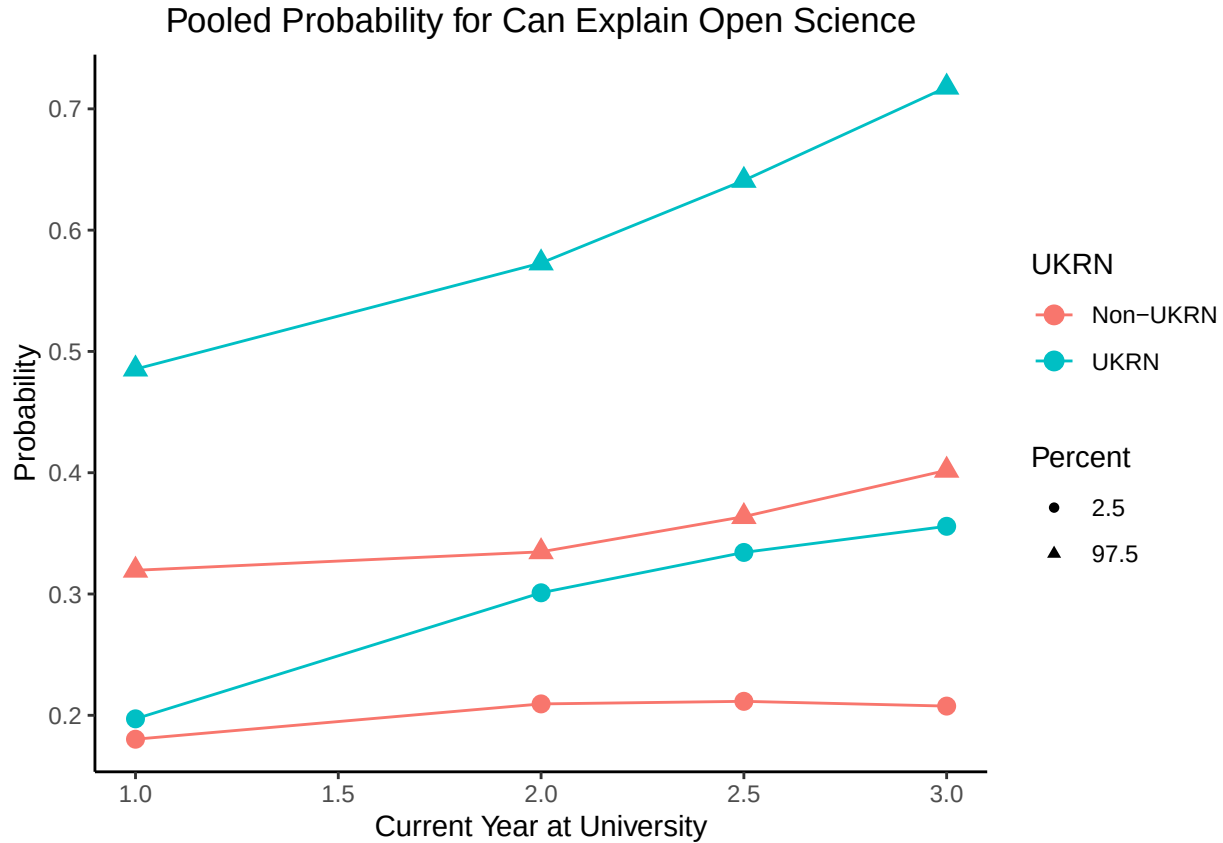
```

Number of participants in the analysis

Total N = 679
 UKRN N = 220
 Non-UKRN N = 459
 Year 1 N = 308

Year 2 N = 253
 Placement Year N = 31
 Year 4 N = 87

The output shows that all beta credibility intervals do cross 0. No significant results.



Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no). The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           2.12    0.01 0.45    1.29    1.80    2.09    2.41    3.05  2041
## sigma        0.66    0.00 0.24    0.29    0.49    0.63    0.80    1.22  2993
## alpha[1]     1.87    0.01 0.64    0.65    1.45    1.85    2.28    3.18  3892
## alpha[2]     2.63    0.01 0.84    1.23    2.04    2.54    3.12    4.50  3235
## alpha[3]     1.86    0.01 0.64    0.63    1.44    1.84    2.26    3.16  3814
## alpha[4]     1.12    0.01 0.33    0.47    0.89    1.12    1.34    1.78  3334
## alpha[5]     2.24    0.01 0.70    1.03    1.76    2.19    2.67    3.77  4098
## alpha[6]     2.20    0.01 0.64    1.05    1.76    2.16    2.60    3.57  3767
```

```

## alpha[7]      2.26      0.01 0.58      1.19      1.86      2.23      2.62      3.47      3127
## alpha[8]      2.27      0.01 0.60      1.17      1.86      2.25      2.65      3.53      3324
## alpha[9]      1.22      0.01 0.38      0.48      0.97      1.22      1.47      1.97      5739
## alpha[10]     1.99      0.01 0.56      0.96      1.61      1.97      2.35      3.14      3799
## alpha[11]     2.56      0.01 0.79      1.24      2.00      2.48      3.04      4.31      3504
## alpha[12]     2.51      0.01 0.80      1.17      1.95      2.43      2.97      4.28      3443
## alpha[13]     2.68      0.01 0.80      1.34      2.14      2.60      3.14      4.47      3186
## alpha[14]     2.21      0.01 0.69      1.00      1.73      2.16      2.63      3.70      4206
## alpha[15]     2.29      0.01 0.60      1.20      1.88      2.26      2.68      3.55      3165
## alpha[16]     1.99      0.01 0.63      0.84      1.57      1.96      2.39      3.30      5074
## alpha[17]     1.74      0.01 0.58      0.62      1.35      1.73      2.11      2.91      4710
## alpha[18]     2.33      0.01 0.69      1.12      1.86      2.27      2.75      3.84      3849
## alpha[19]     2.29      0.01 0.65      1.11      1.83      2.26      2.71      3.66      3354
## beta1         -1.00      0.01 0.84     -2.65     -1.55     -0.98     -0.44      0.62      4541
## beta2         -0.08      0.00 0.19     -0.45     -0.21     -0.08      0.05      0.30      2376
## beta3          0.61      0.01 0.38     -0.11      0.35      0.61      0.86      1.36      5220
## lp__          -299.25      0.11 5.45    -310.57    -302.77    -299.10    -295.58    -288.94      2457
##
## Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## beta1         1
## beta2         1
## beta3         1
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 13:15:05 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

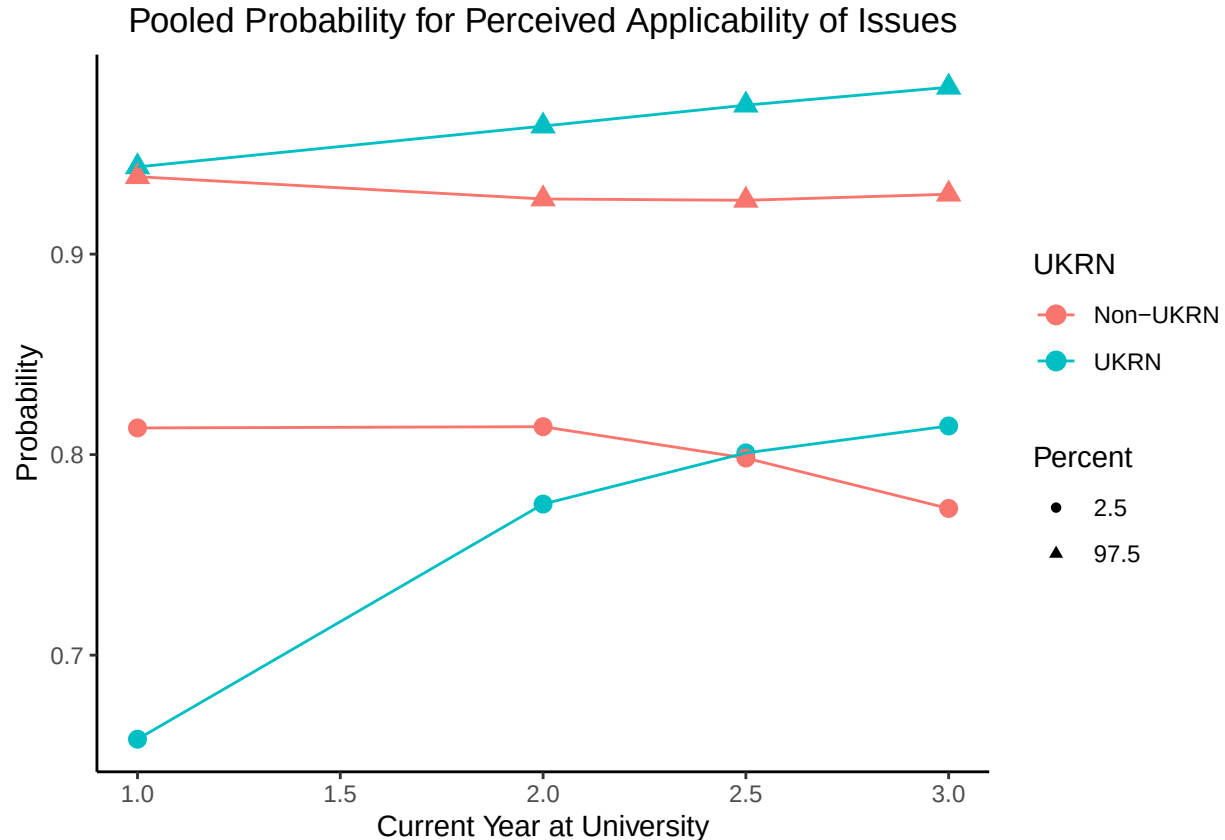
```

Number of participants in the analysis

Total N = 679
 UKRN N = 220
 Non-UKRN N = 459
 Year 1 N = 308

Year 2 N = 253
 Placement Year N = 31
 Year 4 N = 87

The output shows that all beta credibility intervals do cross 0. No significant results.



Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff
## mu           1.40     0.00 0.07   1.26  1.36   1.40   1.45   1.55 2065
## sigma        0.09     0.00 0.03   0.04  0.07   0.09   0.11   0.17 3302
## alpha[1]     1.41     0.00 0.10   1.21  1.35   1.41   1.48   1.62 3212
## alpha[2]     1.41     0.00 0.11   1.20  1.34   1.41   1.48   1.63 3639
## alpha[3]     1.38     0.00 0.10   1.18  1.31   1.38   1.45   1.58 3174
## alpha[4]     1.53     0.00 0.07   1.39  1.48   1.53   1.58   1.66 2948
```

```

## alpha[5]      1.46      0.00 0.11      1.26      1.38      1.45      1.53      1.68 4286
## alpha[6]      1.35      0.00 0.10      1.14      1.28      1.35      1.42      1.54 3469
## alpha[7]      1.35      0.00 0.09      1.16      1.29      1.35      1.41      1.53 2987
## alpha[8]      1.41      0.00 0.09      1.22      1.35      1.41      1.48      1.60 3314
## alpha[9]      1.29      0.00 0.09      1.12      1.23      1.29      1.35      1.45 3258
## alpha[10]     1.37      0.00 0.09      1.18      1.31      1.38      1.44      1.55 3434
## alpha[11]     1.41      0.00 0.10      1.21      1.34      1.41      1.47      1.61 4174
## alpha[12]     1.48      0.00 0.11      1.28      1.41      1.48      1.55      1.72 4067
## alpha[13]     1.39      0.00 0.10      1.19      1.32      1.39      1.46      1.58 3665
## alpha[14]     1.43      0.00 0.10      1.23      1.37      1.43      1.50      1.65 4102
## alpha[15]     1.39      0.00 0.09      1.20      1.33      1.39      1.45      1.57 3300
## alpha[16]     1.40      0.00 0.10      1.19      1.33      1.40      1.47      1.60 3827
## alpha[17]     1.40      0.00 0.10      1.20      1.33      1.40      1.47      1.60 3715
## alpha[18]     1.39      0.00 0.10      1.19      1.32      1.39      1.46      1.59 4008
## alpha[19]     1.40      0.00 0.10      1.20      1.33      1.40      1.46      1.59 3341
## beta1         0.05      0.00 0.15     -0.24     -0.05      0.05      0.15      0.35 4396
## beta2         0.05      0.00 0.04     -0.02      0.03      0.05      0.08      0.12 2222
## beta3         0.09      0.00 0.07     -0.05      0.04      0.08      0.13      0.22 4714
## lp__          1681.16    0.11 5.64 1669.97 1677.49 1681.17 1684.88 1692.23 2448
##              Rhat
## mu            1
## sigma         1
## alpha[1]      1
## alpha[2]      1
## alpha[3]      1
## alpha[4]      1
## alpha[5]      1
## alpha[6]      1
## alpha[7]      1
## alpha[8]      1
## alpha[9]      1
## alpha[10]     1
## alpha[11]     1
## alpha[12]     1
## alpha[13]     1
## alpha[14]     1
## alpha[15]     1
## alpha[16]     1
## alpha[17]     1
## alpha[18]     1
## alpha[19]     1
## beta1         1
## beta2         1
## beta3         1
## lp__          1
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 15:05:42 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 696

UKRN N = 222

Non-UKRN N = 474

Year 1 N = 316

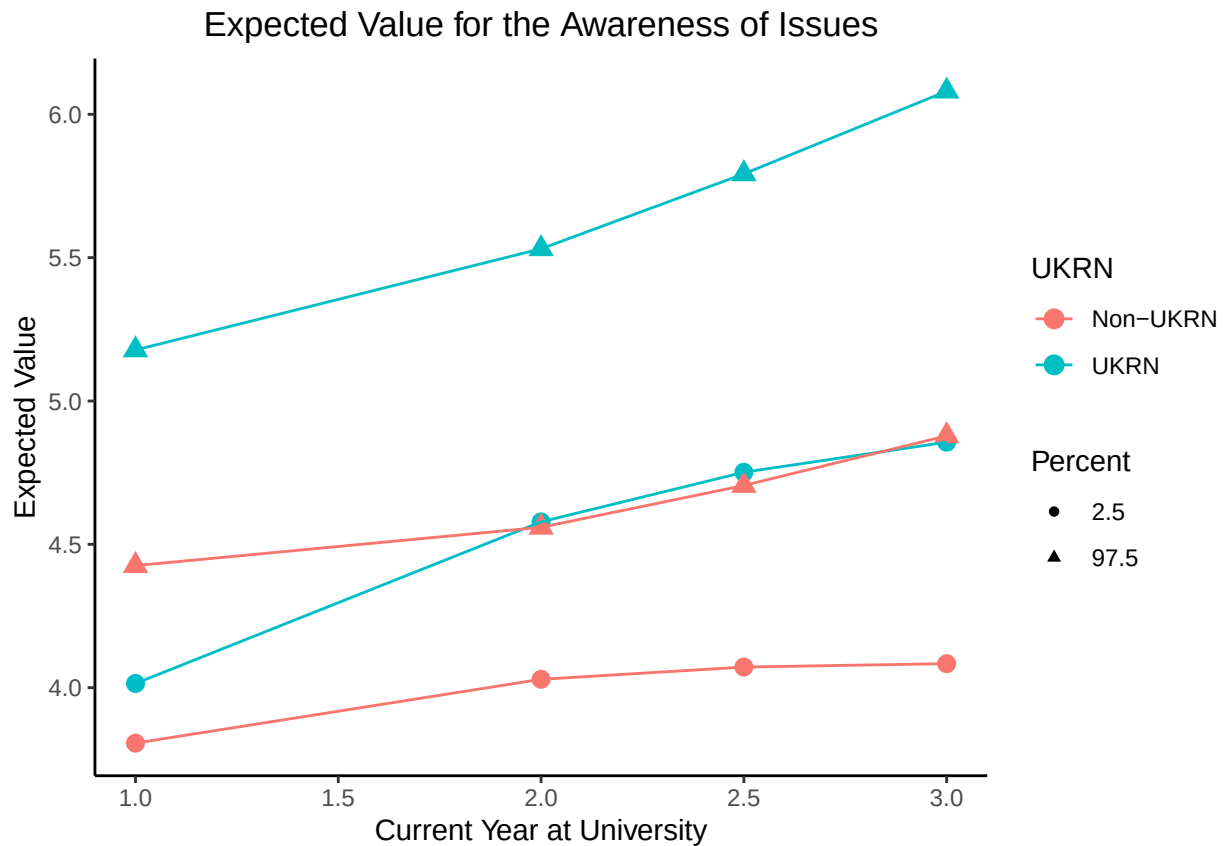
Year 2 N = 258

Placement Year N = 31

Year 4 N = 91

Var1	Freq
0	18
1	39
2	57
3	110
4	135
5	114
6	88
7	59
8	76

The output shows that all beta credibility intervals do cross 0. No significant results.



Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was at a university that was affiliated with the UKRN

* Whether the participant was in their first, second, placement or third year

* Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean    sd    2.5%    25%    50%    75%    97.5% n_eff
## mu            0.69     0.00  0.08    0.53    0.64    0.69    0.75    0.86   328
## sigma         0.08     0.00  0.05    0.01    0.03    0.06    0.10    0.21  432
## alpha[1]      0.70     0.00  0.11    0.48    0.63    0.70    0.76    0.90  569
## alpha[2]      0.68     0.00  0.11    0.44    0.62    0.69    0.76    0.90  639
## alpha[3]      0.70     0.00  0.11    0.48    0.63    0.70    0.77    0.91  604
## alpha[4]      0.69     0.00  0.08    0.54    0.64    0.69    0.75    0.85  322
## alpha[5]      0.77     0.00  0.13    0.56    0.68    0.75    0.83    1.07  828
## alpha[6]      0.74     0.00  0.11    0.54    0.66    0.73    0.81    0.98  722
## alpha[7]      0.68     0.00  0.10    0.48    0.62    0.69    0.75    0.87  441
## alpha[8]      0.74     0.00  0.10    0.55    0.67    0.73    0.80    0.96  673
## alpha[9]      0.70     0.00  0.09    0.53    0.64    0.70    0.76    0.87  446
## alpha[10]     0.70     0.00  0.10    0.50    0.63    0.70    0.76    0.89  479
## alpha[11]     0.70     0.00  0.11    0.48    0.63    0.70    0.77    0.92  663
## alpha[12]     0.62     0.01  0.14    0.29    0.55    0.64    0.71    0.84  473
## alpha[13]     0.72     0.00  0.11    0.52    0.65    0.72    0.79    0.95  648
## alpha[14]     0.62     0.01  0.14    0.29    0.55    0.64    0.71    0.84  478
## alpha[15]     0.68     0.00  0.10    0.47    0.61    0.68    0.75    0.87  434
## alpha[16]     0.66     0.00  0.12    0.40    0.60    0.67    0.74    0.86  552
## alpha[17]     0.70     0.00  0.11    0.48    0.63    0.70    0.77    0.90  643
## alpha[18]     0.66     0.01  0.11    0.41    0.59    0.67    0.74    0.86  463
## alpha[19]     0.70     0.00  0.11    0.49    0.63    0.70    0.77    0.90  598
## beta1         -0.31    0.01  0.19   -0.68   -0.43   -0.31   -0.17    0.07  710
## beta2          0.09    0.00  0.04    0.01    0.06    0.09    0.12    0.17  309
## beta3          0.05    0.00  0.09   -0.13   -0.02    0.05    0.11    0.23  637
## lp__          -281.70    0.77 11.47 -300.66 -290.52 -283.01 -273.40 -258.66 221
##
##               Rhat
## mu            1.02
## sigma         1.01
## alpha[1]      1.01
## alpha[2]      1.01
## alpha[3]      1.01
## alpha[4]      1.02
## alpha[5]      1.00
## alpha[6]      1.01
## alpha[7]      1.02
## alpha[8]      1.01
## alpha[9]      1.02
## alpha[10]     1.02
## alpha[11]     1.01
## alpha[12]     1.02
## alpha[13]     1.01
## alpha[14]     1.02
## alpha[15]     1.02
## alpha[16]     1.02
## alpha[17]     1.01
## alpha[18]     1.02
```

```
## alpha[19] 1.02
## beta1     1.00
## beta2     1.02
## beta3     1.00
## lp__      1.02
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 16:06:20 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 696
 UKRN N = 222
 Non-UKRN N = 474
 Year 1 N = 316
 Year 2 N = 258
 Placement Year N = 31
 Year 4 N = 91

Var1	Freq
0	156
1	110
2	149
3	130
4	73
5	52
6	15
7	10
8	1

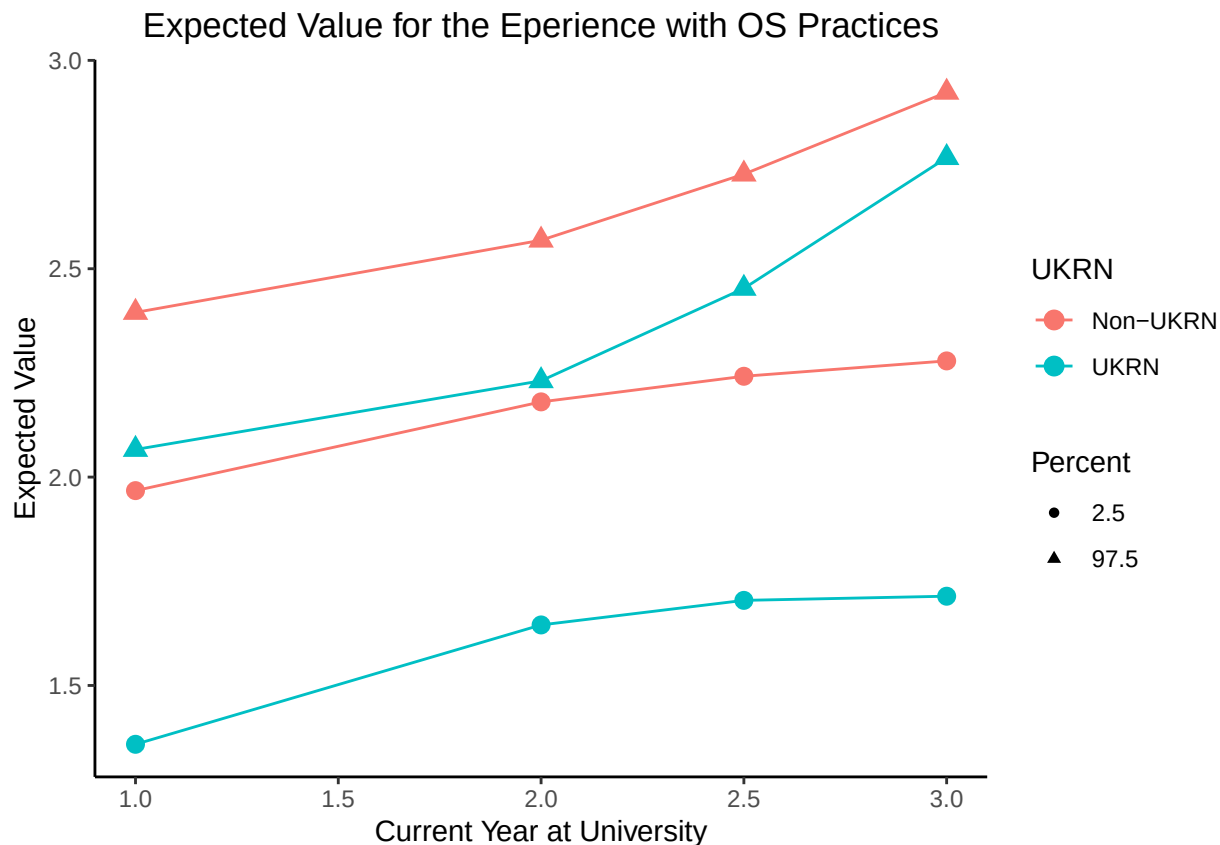
The output shows that all beta credibility intervals cross 0.

The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0050948

97.5% = 0.1664053

The interpretation is that students are more likely to have experience with open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.



Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean  sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu         0.48    0.00 0.05   0.39  0.45  0.48  0.51  0.58  2138   1
## sigma      0.37    0.00 0.01   0.35  0.36  0.37  0.37  0.39 11993   1
## sigma_a    0.07    0.00 0.03   0.02  0.05  0.06  0.08  0.13  1643   1
## alpha[1]   0.42    0.00 0.07   0.28  0.38  0.43  0.47  0.56  3965   1
## alpha[2]   0.50    0.00 0.08   0.35  0.45  0.50  0.55  0.66  3584   1
## alpha[3]   0.52    0.00 0.07   0.38  0.47  0.52  0.57  0.68  3001   1
## alpha[4]   0.43    0.00 0.05   0.34  0.40  0.43  0.46  0.52  3416   1
## alpha[5]   0.55    0.00 0.09   0.40  0.49  0.55  0.61  0.75  2444   1
## alpha[6]   0.52    0.00 0.07   0.38  0.47  0.51  0.56  0.67  2896   1
## alpha[7]   0.49    0.00 0.06   0.37  0.44  0.48  0.53  0.62  3222   1
## alpha[8]   0.48    0.00 0.06   0.36  0.44  0.48  0.52  0.62  3665   1
```

```

## alpha[9]      0.45      0.00 0.05   0.34   0.41   0.45   0.48   0.55  4718   1
## alpha[10]     0.48      0.00 0.06   0.36   0.44   0.48   0.52   0.61  3717   1
## alpha[11]     0.46      0.00 0.07   0.32   0.41   0.46   0.51   0.60  4794   1
## alpha[12]     0.55      0.00 0.09   0.40   0.48   0.54   0.60   0.74  2688   1
## alpha[13]     0.44      0.00 0.07   0.30   0.40   0.44   0.49   0.58  4618   1
## alpha[14]     0.48      0.00 0.07   0.34   0.43   0.48   0.53   0.63  4296   1
## alpha[15]     0.47      0.00 0.06   0.35   0.43   0.47   0.52   0.60  3552   1
## alpha[16]     0.46      0.00 0.07   0.31   0.41   0.46   0.50   0.60  4758   1
## alpha[17]     0.43      0.00 0.07   0.28   0.38   0.43   0.48   0.56  4664   1
## alpha[18]     0.50      0.00 0.07   0.37   0.45   0.50   0.55   0.65  3279   1
## alpha[19]     0.52      0.00 0.07   0.38   0.46   0.51   0.56   0.67  2961   1
## beta1         -0.09      0.00 0.10  -0.29  -0.16  -0.09  -0.02   0.11  4722   1
## beta2          0.05      0.00 0.02   0.00   0.03   0.05   0.07   0.10  2492   1
## beta3          0.07      0.00 0.05  -0.02   0.04   0.07   0.10   0.16  5105   1
## lp__          390.41      0.24 6.84 378.52 385.88 389.94 394.15 406.07   805   1
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 17:09:33 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 690

UKRN N = 221

Non-UKRN N = 469

Year 1 N = 311

Year 2 N = 257

Placement Year N = 31

Year 4 N = 91

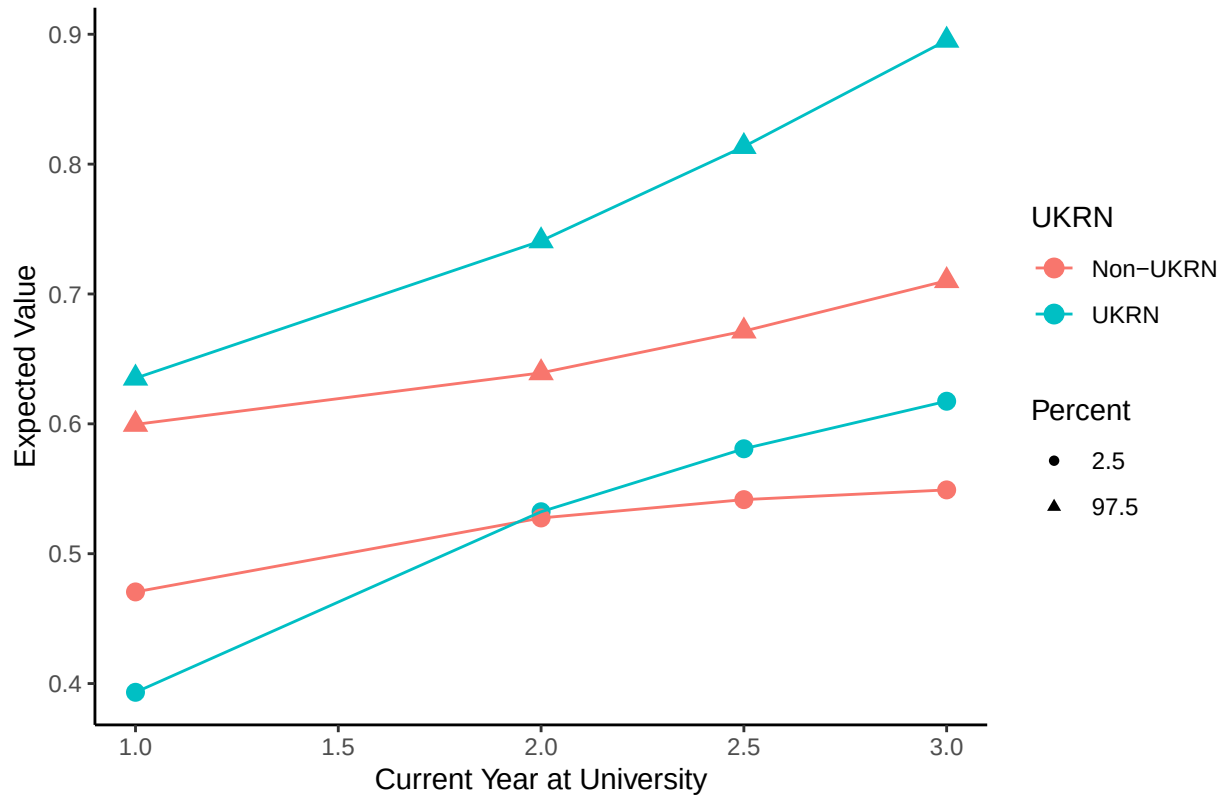
The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0013946

97.5% = 0.0956731

The interpretation is that students are more likely to have positive perceptions of open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.

Expected Value for positive Conceptual Perceptions of OS Practices



Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

- * Whether the participant was at a university that was affiliated with the UKRN
- * Whether the participant was in their first, second, placement or third year
- * Interaction between the two variables above

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu           0.10    0.00  0.06 -0.02  0.06  0.10  0.14  0.23  1200 1.01
## sigma        0.50    0.00  0.01  0.47  0.49  0.50  0.51  0.53  3517 1.00
## sigma_a      0.06    0.00  0.03  0.01  0.03  0.05  0.08  0.14   443 1.02
## alpha[1]     0.06    0.00  0.08 -0.11  0.01  0.07  0.12  0.22  1635 1.00
## alpha[2]     0.11    0.00  0.09 -0.06  0.05  0.11  0.16  0.29  2103 1.00
## alpha[3]     0.13    0.00  0.08 -0.03  0.07  0.12  0.18  0.30  1622 1.00
## alpha[4]     0.08    0.00  0.06 -0.05  0.04  0.08  0.12  0.20  1304 1.01
## alpha[5]     0.15    0.00  0.10 -0.01  0.09  0.14  0.21  0.37  1758 1.01
## alpha[6]     0.12    0.00  0.08 -0.03  0.07  0.12  0.18  0.30  1832 1.00
## alpha[7]     0.10    0.00  0.08 -0.05  0.05  0.10  0.15  0.26  1719 1.00
## alpha[8]     0.11    0.00  0.08 -0.04  0.06  0.11  0.16  0.28  1757 1.00
## alpha[9]     0.12    0.00  0.07 -0.01  0.07  0.12  0.16  0.26  1651 1.00
```

```
## alpha[10] 0.09 0.00 0.08 -0.06 0.04 0.09 0.14 0.25 1850 1.00
## alpha[11] 0.10 0.00 0.08 -0.07 0.05 0.10 0.15 0.27 2130 1.00
## alpha[12] 0.15 0.00 0.10 -0.01 0.09 0.14 0.21 0.38 1765 1.01
## alpha[13] 0.05 0.00 0.09 -0.15 -0.01 0.06 0.11 0.21 1562 1.01
## alpha[14] 0.07 0.00 0.09 -0.12 0.02 0.07 0.13 0.23 1994 1.00
## alpha[15] 0.08 0.00 0.08 -0.09 0.03 0.08 0.13 0.24 1579 1.00
## alpha[16] 0.11 0.00 0.08 -0.05 0.05 0.11 0.16 0.28 2040 1.00
## alpha[17] 0.08 0.00 0.08 -0.09 0.03 0.09 0.14 0.25 2038 1.00
## alpha[18] 0.09 0.00 0.08 -0.08 0.04 0.09 0.14 0.25 2045 1.00
## alpha[19] 0.11 0.00 0.08 -0.05 0.05 0.11 0.16 0.28 1756 1.00
## beta1 0.07 0.00 0.13 -0.19 -0.02 0.07 0.16 0.33 1944 1.00
## beta2 0.14 0.00 0.03 0.08 0.12 0.14 0.16 0.20 1259 1.01
## beta3 0.01 0.00 0.06 -0.12 -0.04 0.01 0.05 0.13 1928 1.00
## lp__ 179.47 0.88 10.46 162.71 172.00 177.94 185.57 204.12 142 1.04
##
## Samples were drawn using NUTS(diag_e) at Wed Sep 25 17:48:05 2024.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 682

UKRN N = 221

Non-UKRN N = 461

Year 1 N = 308

Year 2 N = 255

Placement Year N = 31

Year 4 N = 88

The output shows that beta credibility intervals do not cross 0 only for the effect of year of study.

2.5% = 0.0760839

97.5% = 0.203985

The interpretation is that students are more likely to have positive perceptions of open science practices in the later years of education regardless of whether they are in a UKRN or non-UKRN university.

Expected Value for positive Conceptual Perceptions of OS Practices

